Aristeidis Sionakidis, Ph.D.

Research Associate in Machine Learning

sionaris | Aristeidis Sionakidis | @sionakidis
Personal webpage | as3582@cam.ac.uk | Google Scholar

I am a Research Associate in Machine Learning based at the Precision Breast Cancer Institute (Department of Oncology) at the University of Cambridge. I apply multi-modal and multi-omic learning techniques to predict response and survival in breast cancer treatment. I am interested in interactions between data modalities, data fusion approaches for efficient representation learning, and model interpretability and explainability.

Key skills: Python (PyTorch, scikit-learn) | R (caret, DESeq2, edgeR) | Linux/Bash | working with an HPC | Nextflow

Employment

Research Associate in Machine Learning. 02/2024-Present
University of Cambridge, Department of Oncology, Precision Breast Cancer Institute
Role: Multi-modal integration (multi-omic, clinical and imaging data) in breast cancer research.

Advisor in Bioinformatics and Data Analysis. 04/2021-04/2022
iGEM Thessaloniki, Greece
Role: Early-diagnosis biomarkers for precision medicine interventions in Pancreatic Ductal Adenocarcinoma (PDAC).

Education

Ph.D., University of Edinburgh. MRC DTP in Precision Medicine. 09/2020-03/2024
Thesis: *Integrated analysis of patient gene expression data for Precision Medicine applications in breast and pancreatic cancer.*
Supervisors: Dr. Jonine Figueroa, Dr. Timothy Cannings, Dr. Andrew H. Sims

M.Sc., University of Glasgow. MSc in Precision Medicine and Pharmacological Innovation. 09/2019-09/2020
Grade: Distinction | **Thesis:** *Creating a patient-centric educational platform for personalised medicine.*
Supervisor: Prof. Sandosh Padmanabhan

B.Sc., Aristotle University of Thessaloniki. Pharmacy. 09/2013-07/2018
Grade: 8.57; UK equivalent: **1st class** | **Thesis:** *Cloning of IL-2 coding sequence via recombinant DNA technology.*
Supervisor: Prof. Lefkothea Papadopoulou

Research Publications

Journal Articles

1. T. Papadopoulou, **A. Sionakidis**, A. Piron, and M. Cnop, "Protocol for detecting causal variants by co-localizing gwas and qtl studies using colocredribbon," *STAR Protocols*, vol. 7, no. 3, p. 104 737, 2026, ISSN: 2666-1667. DOI: https://doi.org/10.1016/j.xpro.2026.104737.

2. **A. Sionakidis**, P. N. Lalagkas, A. Malousi, and I. S. Vizirianakis, "Identification of diagnostic markers of pancreatic ductal adenocarcinoma using transcriptomic tumour and blood sample data," *Clinical and Translational Discovery*, vol. 3, no. 5, Oct. 2023, ISSN: 2768-0622. DOI: 10.1002/ctd2.248.

3. C. du Toit, T. Q. B. Tran, N. Deo, S. Aryal, S. Lip, R. Sykes, I. Manandhar, **A. Sionakidis**, L. Stevenson, H. Pattnaik, S. Alsanosi, M. Kassi, N. Le, M. Rostron, S. Nichol, A. Aman, F. Nawaz, D. Mehta, R. Tummala, L. McCallum, S. Reddy, S. Visweswaran, R. Kashyap, B. Joe, and S. Padmanabhan, "Survey and Evaluation of Hypertension Machine Learning Research," *Journal of the American Heart Association*, vol. 12, no. 9, May 2023, ISSN: 2047-9980. DOI: 10.1161/jaha.122.027896.

4. **A. Sionakidis**, L. McCallum, and S. Padmanabhan, "Unravelling the tangled web of hypertension and cancer," *Clinical Science*, vol. 135, no. 13, pp. 1609–1625, Jul. 2021, ISSN: 1470-8736. DOI: 10.1042/cs20200307.

Preprints

1. P. C. Schouten, M. O. Irfan, Z. Kinsella, **A. Sionakidis**, A. Riddell, J. R. Worley, J. Lay, S. Casford, K. Pinilla, J. Kane, D. Whitehorn, S. Tarantino, A. Dayimu, N. Demiris, H. M. Earl, N. Simidjievski, E. Provenzano, and J. E. Abraham, "Tumour region identification guided scoring (TRIGS) and foundation model-based Tumour Infiltrating

Lymphocyte scoring are prognostic for pathological complete response/event free survival in the triple negative patients in the PARTNER randomized controlled trial," Available from medRxiv, Sep. 2026. DOI: 10.64898/2026.09.15.26363116.

2. **A. Sionakidis**, K. Pinilla Alba, J. Abraham, and N. Simidjievski, "Consensus through diversity: A comprehensive benchmark of multi-omic approaches for precision breast oncology," Available from bioRxiv, 2026. DOI: 10.64898/2026.04.17.719159.

3. **A. Sionakidis**, T. I. Cannings, J. D. Figueroa, and A. H. Sims, "A novel gene signature to predict response to neoadjuvant chemotherapy and endocrine treatment in estrogen receptor-positive breast cancer patients," Available from Research Square, Apr. 2023. DOI: 10.21203/rs.3.rs-2771576/v1.

Manuscripts in preparation

1. A. Papadimitriou-Tsantarliotou, **A. Sionakidis**, C. I. Papagiannopoulos, C. Avgeros, A. Malousi, A. K. Anagnostopoulos, I. Papasotiriou, and I. Vizirianakis, *Uncovering STEAP3 as a Ribosome-Associated Protein upon Erythroid Maturation of Murine Erythroleukemia Cells [**Submitted to:** European Journal of Pharmacology]*, Jul. 2026.

2. **A. Sionakidis**, I. Astreidis, E. Galatou, F. Chatzopoulou, K. Paraskevopoulos, A. Malousi, N. Grigoriadis, G. Vatsellas, I. Dimitriadis, K. Vahtsevanos, and I. S. Vizirianakis, *Assessing the clinical and therapeutic impact of molecular biomarkers in recurrent tongue carcinomas by RNAseq analysis [**To be submitted:** Oral Oncology]*, Sep. 2026.

3. **A. Sionakidis** and N. Simidjievski, *MINGLE (Multi-omic INteGration & anaLysis Environment): a library for automated, interoperable, reproducible multi-omic integration*, Sep. 2026.

Grants and Awards

2024-2027	EXACT Sciences grant (co-Investigator) | £1.1 million
09/2021-03/2023	Bodossaki Foundation scholarship for Ph.D. students
10/2021-03/2023	Onassis Foundation scholarship for Ph.D. students
04/2022	Doreen J. Putrah Cancer Research Foundation Scholar-in-Training Award (AACR 2022)
09/2019-09/2020	Bodossaki Foundation scholarship for M.Sc. students
05/2019	Aristotle University of Thessaloniki Excellence Award
03/2017-11/2017	Aristotle University of Thessaloniki undergraduate student scholarship

Active projects

PARTNER	Computational analysis of the multi-modal data from the PARTNER clinical trial (Investigator). *University of Cambridge*.
SYNERGIA	An international collaboration for a multi-modal breast cancer data repository (Investigator). *University of Cambridge*.
SUSA	Sustainable Healthcare with Digital Health Data Competence. **Tutor/Content developer** in "Gene expression, data repositories and standards" and "Data fetching, preprocessing and differential analysis". *Collaboration with Aristotle University of Thessaloniki, Greece*.
Plasma jet	Meta-analysis of plasma jet treatment transcriptomics in skin cells (Investigator). *Collaboration with Aristotle University of Thessaloniki, Greece*.

Presentations

EACR 2026	EACR Cancer Genomics, Multiomics and Computational Biology Conference 2026 poster, Essen
EBCSS 2023	Edinburgh Breast Cancer Society Symposium (EBCSS) 2023 poster, Edinburgh
MEG-UK 2022	Molecular Epidemiology Group (MEG)-UK Annual Meeting 2022 poster, Edinburgh
EACR 2022	European Association for Cancer Research (EACR) Annual Congress 2022 poster, Seville
AACR 2022	American Association for Cancer Research (AACR) Annual Meeting 2022 poster, New Orleans

Leadership & Professional Development

2026 **Resilient Leadership in Action (RLiA)**
Multi-institutional 3-week course: King's College London, Francis Crick Institute, Imperial College London, University of Cambridge, University of Nottingham

2026 **Innovate and Lead: Leadership for Future Science Pioneers**
Workshop: University of Cambridge, Department of Oncology

Teaching

2026 – **University of Cambridge, Department of Genomic Medicine**
MSt in Genomic Medicine
Lecture: "Somatic Variant Interpretation"

2026 – **University of Cambridge, Department of Genetics**
Centre for Research Informatics Training
Member since Sept 2026

2026 **University of Cambridge, School of Clinical Medicine**
Cambridge University Oncology Society Annual Conference
Workshop: "Machine Learning in Oncology: Practical Applications"

2025 – **Aristotle University of Thessaloniki**
M.Sc. in Precision Medicine – Translational Research and Therapeutics
Lecture: "Bioinformatic tools and applications in -omics technologies"

2024 – **Aristotle University of Thessaloniki, Center for Education and Lifelong Learning**
R4LIFE: R Programming Skills for Life Scientists
Lectures: "Basic Functions & Data Wrangling", "Data Analysis & Visualisation", "Analysis of Genomic Data"

Supervision & Mentoring

2024 – **University of Cambridge, Department of Oncology | Supervisions**
Undergraduate research internships: Supervision of summer interns in Python-based data analysis
Visiting medical students: Supervision and training in Python-based data analysis
Work experience: Supervision of sixth form students undertaking research/work experience

2026 – **University of Cambridge, Department of Genomic Medicine**
MSt in Genomic Medicine
Marking: "Pathology of Cancer" module coursework
Supervision: "Automated Multi-omic data Integration Pipelines with MINGLE" | **MSt thesis project**

2026 – **University of Cambridge, Department of Genetics | Supervisions**
NST Part IB Mathematical & Computational Biology
Undergraduate students across Michaelmas, Lent and Easter Terms

2025 **University of Cambridge, Department of Chemical Engineering | Supervisions**
CEBT IA Cell Biology
Undergraduate students

2025 – **Aristotle University of Thessaloniki | Supervisions**
M.Sc. in Precision Medicine – Translational Research and Therapeutics
Supervisions: M.Sc. rotation research projects

2022–2025 **Aristotle University of Thessaloniki | Mentoring & Advising**
PhD researcher: Aliki Papadimitriou-Tsantarliotou
Thesis title: "Molecular analysis of the ribosomal transcriptome during erythropoiesis and an omics-based approach for pharmacological exploitation in ribosomopathies"
Contribution: Provided methodological and analytical guidance, including **study design**, **data analysis**, **code production and review**, and **interpretation** of results

Skills

Data Analysis	Exploratory analysis | data wrangling | hypothesis testing | high-dimensional data | visualization | dynamic report generation | longitudinal & time-to-event/survival data analysis
Computational Biology	Bioinformatics | Nextflow | Genomics: WGS, SNV, CNV, methylation, RNA, neoantigens | differential analysis | network, pathway & deconvolution analysis
Machine Learning	Deep learning with PyTorch, TensorFlow, keras | scikit-learn, caret | multi-omic & multi-modal learning, unsupervised learning | pipeline automation, Hugging Face, MLOps
Programming	Python, R, R Shiny | Linux/Bash/UNIX | HTML, CSS, JavaScript | MySQL, PostgreSQL | Perl | Cloud computing (AWS)
Typesetting	LaTeX, Markdown
Languages	English, French, Greek

Full list of **skills** and **certificates**

Volunteering

2015-2018 **Member at ATP (Aristotle Team of Pharmacy, NGO)**. Thessaloniki
Managerial and soft skills acquired through organizing academic seminars, conferences and social events

References

Available upon request.